Formula sheet for Midterm Exam 2:

1. **Coordinate Conversion Formulas**

Cylindrical to Rectangular	Spherical to Rectangular	Spherical to Cylindrical
$x = r\,cos(\theta)$ $y = r\,sin(\theta)$ $z = z$ $x^2 + y^2 = r^2$	$x = \rho \sin(\varphi)\,cos(\theta)$ $y = \rho \sin(\varphi)\,sin(\theta)$ $z = \rho cos(\varphi)$ $x^2 + y^2 + z^2 = \rho^2$	$r = \rho\,sin(\varphi)$ $z = \rho \cos(\varphi)$ $\theta = \theta$

2. **Fundamental Theorem of Line integrals**:

$$\int_C \boldsymbol{F} \cdot \boldsymbol{T} ds = f(B) - f(A)$$

where *f is* a potential function of vector field **F**, and A, B are end points of curve *C*.

Math261 Midterm Exam 2

Exam2 topic coverage and related practice problems.

1. Double Integrals

   Evaluation of double integrals over regions in plane using:

   - Iterated integration method. (Practice #1, HW4 #s 1-9)
   - Changing the order of integration technique. (Practice #2-5; HW4 #s 16-22)
   - Polar coordinates for circular regions. (Practice #6-8; HW5 #s 1-5)

2. Triple Integrals
   - Evaluation of triple integrals over regions in space using:
     - ✓ Iterated integration method. (Practice #9, 11; HW4 #s 10-15)
     - ✓ Changing the order of integration technique. (Practice #10, 12; HW4 #s 23-27)
   - Set up triple integrals in different coordinates: Rectangular coordinates, Cylindrical coordinates, Spherical coordinates. (Practice #13-23)
   - Application of triple integral to find *volume* or *mass* of a solid. (Practice #24-25; HW 5 #s 6-18)

3. Line integrals of **scalar** functions
   - Evaluation of $\int_C f\,ds$ along a curve C. (Practice #28, 30; HW6 #s 4-6)
   - Applications (Practice #32-34; HW6 #s 1-3; 7-10):
     - ✓ Set up a line integral that represents the *length* of curve.
     - ✓ Set up a line integral that represents the *mass* of curve with given density function.

4. Line integrals of **vector** functions
   - Evaluation of $\int_C \boldsymbol{F}\,d\boldsymbol{r}$ along a curve C (Practice #29, 31; HW6 #s 15-18)
   - Applications: (Practice #35-41; HW6 #s 11-14)
     - ✓ Find work done by a force field $\boldsymbol{F}$ along a parametric curve **r**(t).

   You are expected to know how to set up parametric equations $\boldsymbol{r}(t)$ for circles, ellipses, line segments/triangles, and curves given by $f(x)$ (Practice # 39-41; HW6 #s 14-18).

5. Conservative vector fields

- Determine whether a 2D vector field is conservative. If it is, find a potential function (Practice #42-47; HW6 #s 19-22).
- Use the Fundamental Theorem of Line Integral (FTOL) to evaluate line integrals (Practice #48-50; HW6 #s 23-28)

  Step1: determine the conservativeness. Step2: find its potential function.

  Step3: plug endpoints into potential function and subtract them.

Solution Exam2 Practice

1. $\frac{e-1}{2}\ln 2$
2. $-\frac{1}{\pi}\sin(\pi^2)+\pi$
3. $1/2(e-1)$
4. $\sin(2/3)$
5. $\frac{1}{4}(e^{16}-e)$
6. $\pi/2$
7. $\pi/3$
8. $\frac{\pi}{2}(1-\cos(16))$
9. 0
10. $(-1/2)(\cos(1)-\cos(0))$
11. $1/4\sin(1)(e-1)$
12. $(2/9)(e-2)$
13. $\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_{\sqrt{x^2+y^2}}^{\sqrt{36-x^2-y^2}} 1\,dzdydx$
14. $\int_0^{2\pi}\int_0^1\int_r^{\sqrt{36-r^2}} r\,dzdrd\theta$
15. $\int_0^{2\pi}\int_{\pi/4}^{\pi/2}\int_0^1(\rho\cos(\phi))^2\rho^2\sin(\phi)d\rho d\phi d\theta$
16. $\int_0^{1/\sqrt{2}}\int_0^{2\pi}\int_z^{\sqrt{1-z^2}} z^2 r dr d\theta dz$ or $\int_0^{2\pi}\int_0^{1/\sqrt{2}}\int_0^r z^2 r dz dr d\theta+\int_0^{2\pi}\int_{1/\sqrt{2}}^1\int_0^{\sqrt{1-r^2}} z^2 r dz dr d\theta$
17. (a) $\int_0^{2\pi}\int_0^{\pi/4}\int_0^{3\sec\phi}\rho^2\sin(\phi)d\rho d\phi d\theta$

    (b) $\int_0^{2\pi}\int_0^3\int_r^3 r dz dr d\theta$

    (c) $\int_{-3}^{3}\int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}}\int_{\sqrt{x^2+y^2}}^{3} dzdydx$

18. (a) $\int_0^{2\pi}\int_0^{\pi/3}\int_{\sec\phi}^{2}\rho^2\sin(\phi)d\rho d\phi d\theta$
    (b) $\int_0^{2\pi}\int_0^{\sqrt{3}}\int_1^{\sqrt{4-r^2}} rdzdrd\theta$
    (c) $\int_{-\sqrt{3}}^{\sqrt{3}}\int_{-\sqrt{3-x^2}}^{\sqrt{3-x^2}}\int_1^{\sqrt{4-x^2-y^2}} dzdydx$

19. (a) $\int_0^{2\pi}\int_{\pi/4}^{\pi/2}\int_0^{\csc\phi}\rho^2\sin(\phi)d\rho d\phi s\theta$
    (b) $\int_0^{2\pi}\int_0^{1}\int_0^{r} rdzdrd\theta$
    (c) $\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_0^{\sqrt{x^2+y^2}} dzdydx$

20. $\int_0^{2\pi}\int_0^{1/\sqrt{2}}\int_r^{\sqrt{1-r^2}} rdzdrd\theta$

21. $\int_0^{2\pi}\int_0^{\pi/2}\int_0^{1}\rho^2\sin\phi d\rho d\phi d\theta$

22. $\int_1^5\int_0^{2\pi}\int_0^{z} rdrd\theta dz$

23. $\int_{-1}^{1}\int_0^{2\pi}\int_0^{\sqrt{9-z^2}} rdrd\theta dz$

24. $\int_{-1}^{0}\int_0^{2\pi}\int_0^{\sqrt{100-z^2}} r^3drd\theta dz$

25. $\int_0^{2\pi}\int_0^{2}\int_{-r}^{r} r^3\sin^2(\theta)\cos^2(\theta)dzdrd\theta$

26. $\int_0^{2\pi}\int_0^{2}\int_{r^2-6}^{8-2r^2} rdzdrd\theta$

27. $\int_0^{2\pi}\int_1^{2}\int_0^{r^2+1} rdzdrd\theta$

28. 0

29. -2/3

30. $\frac{\sqrt{2}}{3}\sin^3(\pi/2)$

31. $\pi/2+(1/3)(\pi/2)^3$

32. (a) $\int_0^1\sqrt{4t^2+25+e^{2t}}dt$; (b) $\int_0^1 t^4\sqrt{4t^2+25+e^{2t}}dt$

33. (a) $\int_0^{2\pi}\sqrt{4\cos^2(t)+4\sin^2(t)+1}dt = \int_0^{2\pi}\sqrt{4+1}dt = \sqrt{5}\int_0^{2\pi}dt$; (b) $\sqrt{5}\int_0^{2\pi}(4-t)dt$

34. The parameterization is $x(t)=2\cos(t), y(t)=2\sin(t)$
    (a) $\int_0^{\pi}\sqrt{4\sin^2(t)+4\cos^2(t)}dt = 2\int_0^{\pi}dt = 2\pi$
    (b) $\int_0^{\pi}(2\cos(t))^2 2\sin(t)\sqrt{4\sin^2(t)+4\cos^2(t)}dt = 16\int_0^{\pi}\cos^2(t)\sin(t)dt$

35. $\int_0^5\langle t^2,2\rangle\cdot\langle 1,e^t\rangle dt = \int_0^5 t^2+2e^t dt = \frac{1}{3}t^3+2e^t|_0^5 = \frac{1}{3}(5)^3+2e^5-2$

36. $-2/3$
37. $1/3$
38. $1/5 + 1/2$
39. 0
40. 8π
41. 1
42. It is not conservative
43. $xy^2 + e^x + 1/2y^2$
44. It is not conservative
45. $ye^x + e^y$
46. $x^2e^{-y} + y^2$
47. $x\sin(y) + e^y$
48. $\frac{2}{3}$
49. 2
50. $\sin(1)$
51. Choices b, d, f, and g.

Math 261: Exam2 Practice Problems

1. Evaluate the integral $\int_0^1 \int_0^1 \frac{xe^y}{1+x^2} dxdy$
2. Evaluate the integral $\int_0^\pi \int_0^\pi y \sin(xy) dydx$
3. Evaluate the integral $\int_0^1 \int_y^1 e^{x^2} dxdy$
4. Evaluate the integral $\int_0^1 \int_0^{\sqrt{y}} \cos(x - \frac{1}{3}x^3) dxdy$
5. Evaluate the integral $\int_0^1 \int_1^2 e^{x^4} dxdy + \int_1^8 \int_{y^{1/3}}^2 e^{x^4} dxdy$
6. Evaluate the integral $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} dydx$
7. Evaluate the integral $\int_{-1}^1 \int_{-\sqrt{1-y^2}}^0 \sqrt{x^2+y^2} dxdy$
8. Evaluate the integral $\int_{-4}^4 \int_0^{\sqrt{16-y^2}} \sin(x^2+y^2) dxdy$
9. Evaluate the integral $\int_{-1}^1 \int_0^2 \int_0^{xy} x^2 e^{y^2} dzdxdy$
10. Evaluate the integral $\int_0^1 \int_0^{2z} \int_{y/2}^z z \cos(x^2) dxdydz$
11. Evaluate the integral $\int_0^1 \int_0^1 \int_{xz}^{2xz} \cos(z^2) e^{x^2} dydxdz$
12. Evaluate the integral $\int_0^1 \int_0^z \int_{y^2}^{z^2} z^2 e^{x^{3/2}} dxdydz$
13. Set up, but do not solve, a triple integral which represents the volume of the region above the cone $z = \sqrt{x^2+y^2}$, within the cylinder $x^2+y^2 = 1$, and below the sphere $x^2+y^2+z^2 = 36$, using cartesian coordinates
14. Repeat the previous problem, but do it in cylindrical coordinates.
15. Suppose that the density of the solid that lies within the sphere $x^2+y^2+z^2 = 1$, below the cone $z = \sqrt{x^2+y^2}$, and above the plane $z = 0$, is given by the function $f(x,y,z) = z^2$. Set up, but do not solve, a triple integral representing the mass of this object in spherical coordinates.
16. Repeat the previous problem but do it in cylindrical coordinates.
17. Set up, but do not solve, a triple integral that represents the volume of the solid that is above the cone $z = \sqrt{x^2+y^2}$, and below the plane $z = 3$.
    (a) In spherical coordinates:
    (b) In cylindrical coordinates:

(c) In cartesian coordinates:

18. Set up, but do not solve, a triple integral that represents the volume of the solid that is within the sphere $x^2 + y^2 + z^2 = 4$, and above the plane $z = 1$.

    (a) In spherical coordinates:

    (b) In cylindrical coordinates:

    (c) In cartesian coordinates:

19. Set up, but do not solve, a triple integral that represents the volume of the solid that is inside the cylinder $x^2 + y^2 = 1$, below the cone $z = \sqrt{x^2 + y^2}$, and above the plane $z = 0$.

    (a) In spherical coordinates:

    (b) In cylindrical coordinates:

    (c) In cartesian coordinates:

20. Rewrite the integral $\int_0^{2\pi} \int_0^{\pi/4} \int_0^1 \rho^2 \sin(\phi)\, d\rho d\phi d\theta$ using cylindrical coordinates.

21. Rewrite the integral $\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} 1\, dz dy dx$ using spherical coordinates.

22. Suppose a solid lies within the cone $z = \sqrt{x^2 + y^2}$, above the plane $z = 1$ and below the plane $z = 5$. Set up a triple integral, in any coordinate system, that represents the volume of this solid.

23. Suppose a solid lies within the sphere $x^2 + y^2 + z^2 = 9$, below the plane $z = 1$, and above the plane $z = -1$. Set up a triple integral, in any coordinate system, that represents the volume of this solid.

24. Suppose that the density of the solid that is within the sphere $x^2 + y^2 + z^2 = 100$, below the plane $z = 0$, and above the plane $z = -1$, is given by $\delta(x, y, z) = x^2 + y^2$. Set up a triple integral, in any coordinate system, that represents the **mass** of this solid.

25. Suppose that the density of the solid that lies within the cylinder $x^2 + y^2 = 4$, below the cone $z = \sqrt{x^2 + y^2}$, and above the cone $z = -\sqrt{x^2 + y^2}$, is given by $\delta(x, y, z) = x^2 y^2$. Set up a triple integral, in any coordinate system, that represents the **mass** of this solid.

26. Suppose a solid lies within the cylinder $x^2 + y^2 = 4$, above the paraboloid $z = x^2 + y^2 - 6$ and below the paraboloid $z = 8 - 2x^2 - 2y^2$. Set up a triple integral, in any coordinate system, that represents the volume of this solid.

27. Suppose a solid lies outside the cylinder $x^2 + y^2 = 1$, inside the cylinder $x^2 + y^2 = 4$, above the plane $z = 0$ and beneath the paraboloid $z = x^2 + y^2 + 1$. Set up a triple integral, in any coordinate system, that represents the volume of this solid.

28. If C is the curve parameterized by $\mathbf{r}(t) = \langle t, t^2 \rangle$, for $-1 \leq t \leq 1$, and $f(x, y) = x$, compute the integral $\int_C f\, ds$.

29. If C is the curve parameterized by $\mathbf{r}(t) = \langle t, t^2 \rangle$, for $-1 \leq t \leq 1$, and $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$, compute the integral $\int_C \mathbf{F} \cdot d\mathbf{r}$.

30. If C is the curve parameterized by $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$, for $0 \leq t \leq \pi/2$, and $f(x, y, z) = xy^2$, compute the integral $\int_C f \, ds$.

31. If C is the curve parameterized by $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$, for $0 \leq t \leq \pi/2$, and $\mathbf{F} = -y\,\mathbf{i} + x\,\mathbf{j} + z^2\,\mathbf{k}$, compute the integral $\int_C \mathbf{F} \cdot d\mathbf{r}$.

32. Suppose a wire runs along a curve that is parameterized by $\mathbf{r}(t) = \langle t^2, 5t, e^t \rangle$ for $0 \leq t \leq 1$, and has density $\delta(x, y, z) = x^2$ *kg/m*.

    (a) Set up, but do not solve, an integral that represents the *length* of the wire.

    (b) Set up, but do not solve, an integral that represents the *mass* of the wire.

33. Suppose a wire runs along the curve that is parameterized by $\mathbf{r}(t) = \langle 2\cos(t), 2\sin(t), t \rangle$ for $0 \leq t \leq 2\pi$, and has density $\delta(x, y, z) = 4 - z$ *kg/m*.

    (a) Set up, but do not solve, an integral that represents the *length* of the wire.

    (b) Set up, but do not solve, an integral that represents the *mass* of the wire.

34. Suppose a wire runs along a curve given by the top half of the circle $x^2 + y^2 = 4$, with density $\delta(x, y) = x^2 y$ *kg/m*. Parameterize this curve and answer the following questions.

    (a) Set up, but do not solve, an integral that represents the *length* of the wire.

    (b) Set up, but do not solve, an integral that represents the *mass* of the wire.

35. Find the work done by the force field $\mathbf{F}(x, y) = x^2\,\mathbf{i} + 2\,\mathbf{j}$ by moving a particle along the curve $\mathbf{r}(t) = \langle t, e^t \rangle$ for $0 \leq t \leq 5$.

36. Find the work done by the force field $\mathbf{F}(x, y) = y\,\mathbf{i} - x\,\mathbf{j} + z\,\mathbf{k}$ by moving a particle along the curve $\mathbf{r}(t) = \langle t, t^2, t^4 \rangle$ for $-1 \leq t \leq 1$.

37. Find the work done by the force field $\mathbf{F}(x, y) = y^2\,\mathbf{i} + (y - x)\,\mathbf{j}$ by moving a particle along the line segment from $(0, 1)$ to $(1, 0)$.

38. Find the work done by the force field $\mathbf{F}(x, y) = (x^2 y)\,\mathbf{i} + y\,\mathbf{j}$ by moving a particle along the curve $y = x^2$ for $0 \leq x \leq 1$.

39. Find the work done by the force field $\mathbf{F}(x, y) = x\,\mathbf{i} + y\,\mathbf{j}$ by moving a particle once along the ellipse $\frac{x^2}{9} + \frac{y^2}{16} = 1$, in the **clockwise** direction.

40. Find the work done by the force field $\mathbf{F}(x, y) = -y\,\mathbf{i} + x\,\mathbf{j}$ by moving a particle once along the circle $x^2 + y^2 = 4$, in the **counterclockwise** direction.

41. Find the work done by the force field $\mathbf{F}(x, y) = (x^2 - y^2)\mathbf{i} + (xy)\mathbf{j}$ by moving a particle once along the triangle with vertices $(0, 0)$, $(0, 1)$ and $(1, 1)$, in the **counterclockwise** direction.

42. Determine whether or not the vector $\mathbf{F} = y^2\,\mathbf{i} + x^4\,\mathbf{j}$ is conservative. If it is, compute the potential function.

43. Determine whether or not the vector $\mathbf{F} = (y^2 + e^x)\,\mathbf{i} + (2xy + y))\,\mathbf{j}$ is conservative. If it is, find a potential function.

44. Determine whether or not the vector $\mathbf{F} = (yx^3)\,\mathbf{i} + (y^2x^2)\,\mathbf{j}$ is conservative. If it is, find a potential function.

45. Determine whether or not the vector $\mathbf{F} = (ye^x)\,\mathbf{i} + (e^x + e^y)\,\mathbf{j}$ is conservative. If it is, find a potential function.

46. Determine whether or not the vector $\mathbf{F} = (2xe^{-y})\,\mathbf{i} + (2y - x^2e^{-y})\,\mathbf{j}$ is conservative. If it is, find a potential function.

47. Determine whether or not the vector $\mathbf{F} = \sin(y)\,\mathbf{i} + (x\cos(y) + e^y)\,\mathbf{j}$ is conservative. If it is, find a potential function.

48. Let $\mathbf{F} = x^2\,\mathbf{i} + y^2\,\mathbf{j}$. Use the Fundamental Theorem of Line Integrals to evaluate $\int_C \mathbf{F}\cdot\mathbf{T}\,ds$, where C is the portion of the parabola $y = 2x^2$ from (-1, 2) to (1, 2).

49. Let $\mathbf{F} = xy^2\,\mathbf{i} + x^2y\,\mathbf{j}$. Use the Fundamental Theorem of Line Integrals to evaluate $\int_C \mathbf{F}\cdot\mathbf{T}\,ds$, where C is a path from (0, 1) to (2, 1).

50. Let $\mathbf{F} = e^x\sin(y)\,\mathbf{i} + e^x\cos(y)\,\mathbf{j}$. Use the Fundamental Theorem of Line Integrals to evaluate $\int_C \mathbf{F}\cdot\mathbf{T}\,ds$, where C is the portion of the circle $x^2 + y^2 = 1$ from (1, 0) to (0, 1).

51. Which of the following integrals are set up in a way that makes mathematical sense? (There may be multiple correct answers).

    (a) $\int_0^1\int_0^z\int_x^z f(x,y,z)dxdydz$

    (b) $\int_0^1\int_0^z\int_y^z f(x,y,z)dxdydz$

    (c) $\int_0^z\int_0^{z^2}\int_0^x f(x,y,z)dxdydz$

    (d) $\int_0^1\int_{z^2}^z\int_0^y f(x,y,z)dxdydz$

    (e) $\int_0^1\int_{z^2}^z\int_x^1 f(x,y,z)dxdydz$

    (f) $\int_0^1\int_0^1\int_0^{x^2y^2} f(x,y,z)dzdxdy$

    (g) $\int_0^1\int_x^1\int_0^{x+y} f(x,y,z)dzdydx$

    (h) $\int_0^1\int_0^x\int_0^z f(x,y,z)dzdydx$